AE460 A/B
REPORT CONTENT REQUIREMENTS
Fall 2024, REV D, 8/25/24

Usage of the Report Content Requirements (RCR) ensures uniformity of the design report. Where conflict exists between the System Requirements Document (SRD) and the RCR, the SRD takes precedent. Your reports should present the same level of professionalism and general format of the winning reports from the prior AIAA design competitions, which are available in Canvas under "Course Documents." Where the AIAA design competition request for proposal (RFP) differs from these instructions, notify Professor Butler and follow the AIAA guidance.

1 Executive Summary

This section is your creative sales pitch. Think of it as a detailed automotive sales brochure that makes a customer want to buy one, or at a minimum wants to know more.

1.1 Description

1.1.1 Describe the design and why it is best for the design mission. This is the page turner section, no more than two pages long. Remember that a picture is worth a thousand words, so consider using a graphic, perhaps containing key technical information, to introduce your aircraft. See Jane's or on-line sources for examples.

1.1.2 Present a high level Operational View (OV-1) diagram with **key requirements, design missions** and **design features** of the aircraft that meet the requirements. Print on a single 11x17" sheet. You may use 2 sheets if needed to convey your message.

1.2 Status of the Design

Describe the status of the design. Describe any key assumptions, deficiencies or errors that might remain in the design and include a description of the design changes to be made for a next iteration. Details of changes shall be documented in section 2, including the supporting rationale.

2 System Requirements

This section documents the breakdown of the System Requirements Document (SRD) and allocation of the requirements to each of the respective responsible group, i.e. aerodynamics, performance, etc.

2.1 Identification and Allocation

The breakdown and allocation of the requirements shall as follows:

Table 2-1 System Requirements (Example)

SRD Paragraph numl	Requirement (full)	Aerodynamic	Performanc	Propulsio	Stability and Contr	Airframe/Subsysten	Mass Propertie	Require	Desire	Achieve
4.1.4	Provisions for 50 lbs. of baggage are required, in a 24” x 18” x 12” volume.					X	X			
4.10.5	Take off ground run 2500 ft maximum, Configuration WTO, maximum ramp weight less 200lb, accelerate to take off speed and hold that speed for 3s, then rotate	X	X	X	X			<2500 ft	N/A	2600 ft
4.10.4	Maximum speed M=0.85, Configuration WCR, 60% fuel, 35,000 ft	X	X	X	X	X	X	0.85	N/A	0.8
5.2	Phugoid damping ratio				X			>.04	N/A	0.05
	etc.									

Dual requirements may be broken into two or more, if allocations are different, i.e. the weight and volume requirement of 4.1.4 may be re-written to split into 4.1.4a and 4.1.4b, and allocated to different groups. The achieved value has colored cells indicating the result. This helps the reader to focus on the risks of the design and what needs to be further discussed. Colors and text as follows:

Cell Color/Text	Meaning
Red	Not met
Yellow	Not met, but close
Blue	Met
Text	State value attained, if appropriate
TBD	Not yet evaluated or has yet to be documented (use sparingly)

2.2 Design Changes Required

Any **RED** or **YELLOW** requirement shall be documented in this section and have the planned changes to return the requirement to **GREEN**. Include any supporting rationale.

3 Aircraft Description

3.1 Preliminary Sizing (Roskam, Part I)

3.1.1 Preliminary Weight Estimate (Roskam, Part I): include all assumptions, graphs, comparisons of weights with similar aircraft, fuel fractions, preliminary weight breakdown, etc.

3.1.2 Wing Area (S) and Takeoff Thrust (T_{TO}) Estimate (Roskam, Part I): include all assumptions and tables to build the wing loading matching graph. The final graph shall include wing loading and takeoff thrust chosen for you aircraft. Include discussion on why you chose the values.

3.1.3 Landing Gear Loads and Tire Sizing (Roskam, Part IV): include a drawing of the landing gear layout (stick figure acceptable) with loads for each strut. Specify tires chosen for use (include the tire specifications in your discussion).

3.2 Preliminary Configuration (Roskam, Part II, III & IV)

3.2.1 Describe the progression of the design, from the preliminary layout sketches, drawings and the design decisions that made it evolve into the final design. Include comparison of other aircraft in your discussion.

3.2.2 Describe the influence of design requirements on the configuration, include the following in the discussion:

3.2.2.1 Cockpit location

3.2.2.2 Airfoil choices

3.2.2.3 Wing placement

3.2.2.4 Planform, sweep, taper ratio, aspect ratio, dihedral/anhedral

3.2.2.5 Tail sizing and tail placement

3.2.2.6 Control surfaces

3.2.2.7 High lift devices

3.2.2.8 Payload placement

3.2.2.9 Engine placement

3.2.2.10 Landing gear type and placement

3.2.2.11 Any other special features

3.3 Configuration and Drawings

All drawing sheets shall have the team number identified in, or above, the title block, i.e. TEAM X, where X is the team number.

3.3.1 General Arrangement 3-View Drawing

Include a scale 11” x 17” 3-view drawing (front, planform and side view) showing all external features with external payloads, including external fuel tanks. Drawing shall include the following dimensions in inches/degrees, except as noted:

Aircraft Coordinate System (ACS) (0,0,0) location on each view
Overall length (ft)
Overall height from ground (ft), with landing gear down
Overall span, including wing tip tanks, payloads, lights, etc. (ft)
CG location on planform view (FWD and AFT) (include FS location and %mac)
CG location on side view (FWD and AFT) (include WL location)
Landing Gear Down (stick figure only, no gear doors)
Tip back angles and lateral tip over criteria
Pilot/Co-Pilot design eye height location (FS and WL)
Pilot/Co-Pilot look angles (forward and side)
External payload hardpoint locations (BL)
Leading Edge (LE) sweep angle (one panel only, each surface)
Mean Aerodynamic Chord (MAC, one panel only, each surface)
Wing LE MAC location from FS0 (one panel only)
Aircraft aerodynamic center (ac_a) FS location (include %mac)
Moment arm length between the .25% MAC of surfaces (used for tail volume calculation)

3.3.2 Drawing Table

Typically placed on the General Arrangement 3-view, this table is to be placed on a second 11x17 sheet for instructor convenience (readability). All units shall be inches and degrees, unless otherwise specified. Table shall show the following numerical data with additions/deletions as appropriate (**in this order**):

Wing

- Planform Area, S (ft^2)
- Exposed Area (ft^2)
- Wetted Area (ft^2)
- Reference Span, b (ft)
- Apex location, FS, BL and WL (in)
- Root chord (in)
- Tip chord (in)
- MAC (in)
- Taper ratio (λ), Aspect ratio
- Sweep: LE, .25c, max thickness line, .50c, TE
- Incidence angle
- Airfoil section(s) and twist

Vertical, Horizontal and Canards (if applicable)

- Same parameters as wing above
- Horizontal and vertical tail volume ratios

High Lift devices and control surfaces

- Type of high lift device
- % chord and % span
- Area affected of surface (ft^2)
- Area of surface aft of hinge line (ft^2)
- MAC of control surface (in)

Fuselage

- Length (ft)
- Max cross section area (ft^2)
- Fineness ratio
- Wetted area (ft^2)

Canopy

- Length (ft)
- Wetted area (ft^2)

Total airplane wetted area (ft^2)

3.3.3 Inboard Profile

Include a scale 11” x 17” inboard profile showing locations of all equipment, cockpit, structure and systems. May use multiple sheet for clarity of the drawing views.

Show and identify in the drawing at least the following, if installed:

Structure, including spars, ribs, frames, bulkheads, longerons, etc.
Control Surfaces
High lift device(s)
Dive brakes
Landing gear: up and down
Gear doors: up and down
Pilot Seats (human factors)[1]
Crew Station Panels (human factors)[1]
Pilot Stick/Throttle (human factors)[1]
Rudder Pedal (human factors)[1]
Crew, passengers, and seats (human factors)[1]
Passenger doors/steps
Crew Ladder
Baggage
Oxygen system (tank/OBOGS)
Environmental Control System (ECS) (AC and heat)
Other crew/passenger supplied items
External stores
Internal ordnance and ammunition
Other payload
Engines and ducts
Propellers or fans
Fuel tanks and pumps
Avionics and instruments
Tail hook, up and down (if applicable)

[1] Human Factor: These items shall be presents in its own detailed view showing how they meet human factor requirements.

4 Mass Properties

Mass Properties, which include Weight, Center of Gravity (CG), and Moments of Inertia (MOI), shall be provided in Aircraft Coordinate System (ACS) as shown in Figure 4-1 below. The abbreviations FS, BL and WL stand for Frame Station, Water Line and Butt Line respectively. These terms come from a long tradition in ship design and building. (Side note: when discussing Weight and CG, it is typically referred to Weight and Balance)

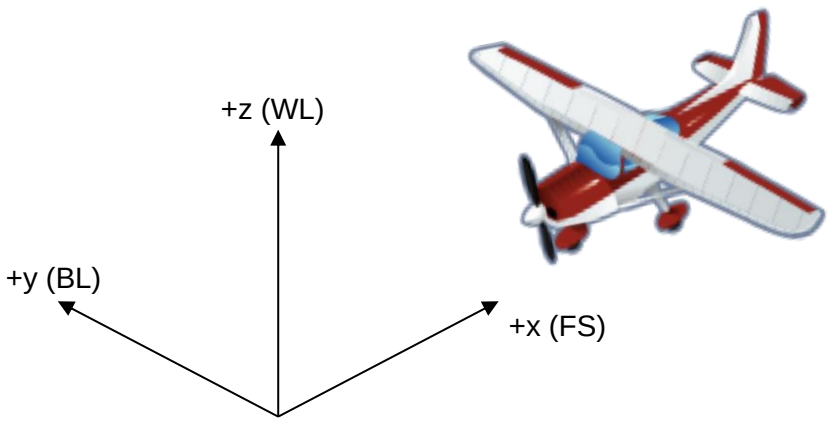

Figure 4-1 Aircraft Coordinate System Convention

4.1 Detailed Weight Estimate

Include all equations used, showing the calculations with inputs and outputs (see Nicolai CH 20, or Roskam V).

Include a weight summary table for both missions in this format:

		Weight (lb)
Operating Empty Weight (W_{OE})		
	Empty Weight (W_E)	
	Trapped Fuel and Oil (W_{tfo})	
	Crew Weight (W_{CREW})	
Fuel (W_F)		
W_{OE} + 60% fuel		
$W_{OE} + W_F$		
Payload Weight (W_{PL})		
$W_{OE} + W_F + W_{PL}$		

4.2 Weights and CG

Include all component weights and locations in ACS to obtain an estimated CG location of the aircraft, see Table 4-1 for example. As the design matures so does this table (See Roskam Part II CH 10 for CG estimation of major components).

Include tables for each configuration, such as CC, FC, WC, etc.

Table 4-1 Mass Properties (example)

		Weight	Distance			Moment		
		W	X	Y	Z	X	Y	Z
Item	Name	lb	in	in	in	lb-in	lb-in	lb-in
	Wing							
	Tail-Horiztonal							
	Tail-Vertical							
	Fuselage							
	SW		lb		SM			
			X_{CG}	Y_{CG}	Z_{CG}			
			in	in	in			
			%mac					

4.3 Longitudinal CG travel

Show the following CG locations in % MAC and distance from Aircraft Coordinate System fixed x-y-z reference (FS) for all configurations, i.e. CC, FC, WC:

Payload	Gear	% Fuel	FS (in)	%MAC
Full Payload	Up	1		
		60		
		100		
	Down	1		
		60		
		100		
No Payload	Up	1		
		60		
		100		
	Down	1		
		60		
		100		

4.4 Moments of Inertia **(AE 460B only)**

Expanding on the Mass Properties table, Moments of Inertia (Ixx, Iyy, Izz and Ixz only) are required for only one flight condition to support the stability control analysis, see the SRD. Start with the mass properties table and add columns to include:

- Radius of gyration or Moments of Inertia (MOI) of each component
- MOI of components transformed and summed at the ACS (0, 0, 0)
- MOI transformed to body axis system about the CG

5 Flight Loads (AE 460B only)

5.1 Using MIL-A-8861 or 14CFR25, depending on the SRD, develop and present the composite V-n Diagrams (n vs. EAS) as follows:

- lightest weight configuration, at sea level
- lightest weight configuration, at highest cruise altitude
- heaviest weight configuration, at sea level
- heaviest weight configuration, at highest cruise altitude

5.2 Calculate aircraft loads (Aerodynamic loads) on main wing only IAW the methodology in NACA TR 921, or Airframe Structural Design (Niu) or equivalent.

6 Structures (AE 460B only)

Using the developed loads perform first order structural sizing of the main spar(s) of the main wing. Use the methodology in Airframe Structural Design (Niu), Analysis & Design of Flight Vehicle Structures (Bruhn) or an equivalent source.

This section is being developed and will be ready for AE460B. The activity will be similar to AE310 final project.

7 Lift, Drag and Moment Estimation

7.1 Airfoil curves Cl vs. α curves

7.1.1 Main Wing Airfoil

7.1.2 Horizontal Stabilizer Airfoil

7.1.3 Vertical Stabilizer Airfoil (if different from horizontal)

7.2 Lift Curves, C_L vs α (for takeoff, climb, landing)

7.2.1 Conditions: Reynolds Number for V = 100 KTAS, Sea Level, 90°F day, Configuration: CC

7.2.2 Takeoff/landing Curve "Build-up"

7.2.2.1 Show all the details to derive the curves for airfoil, wing, flaps and slats. Document all calculations and adjustments made for twist and incidence and horizontal stabilizer contribution. Construction of curves may be done on graph paper by hand (easiest way).

7.2.3 Aircraft C_L vs α curves (takeoff, climb, landing)

The graph simplifies and "cleans up" the curves constructed in section 7.2.2, which shall show the C_L vs α of the entire aircraft for CC, takeoff and landing configurations.

Note: Information on each C_L vs α graph in section 7.2 shall include the following information:

Airfoil identification
Re number
S (reference area)
AR (aspect ratio)
Mach
Sweep at max thickness line
Twist angle
Incidence angle

7.3 C_L vs α, Flight Condition to be analyzed (as required)

7.3.1 See SRD. Flight condition curves will be needed to evaluate various flight conditions, i.e. high-g turns, subsonic and supersonic cruise conditions. Only develop a C_{Lmax} if required. For most cases, only the linear portion of the curve will be required for performance, stability and control purposes.

7.4 Zero Lift Drag Coefficient

Show sample calculations with step by step process for each type of analysis, i.e. wing, fuselage, with each of the air speed regime, i.e. subsonic, transonic, and supersonic as appropriate.

7.4.1 C_{Do} vs. M graph showing the contribution of each component and combined results.

7.4.2 C_{Do} vs. M graph showing only CC, WC & FC, for applicable SRD configurations

7.4.3 Sub Critical Mach Drag Buildup Equivalent Flat Plate Area (ft^2) Summary

Table 7-1 Equivalent Flat Plate Area Summary (example)

	Configuration			
	CC	WC	WTO	WL
Wing				
Horizontal Tail				
Vertical Tail				
Fuselage				
Engine Housing/Nacelle				
Boundary Layer Diverter				
Boattail				
Canopy				
Proturberances				
Fuel Probe				
SUBTOTAL				
External Stores				
Pylons				
SUBTOTAL				
Landing Gear				
Slats				
Flaps				
Hook				
SUBTOTAL				
Dead Engine				
Speed Brakes				
SUBTOTAL				
	CC	WC	WTO	WL
TOTAL				

7.4.4 Sub-critical Mach Drag Buildup C_{Do} Summary

C_{Do}s for Components, based on Wing Planform Area. (Table identical to the above)

7.5 Drag Due to Lift Factor (K, K', K'')

Show sample calculations with step by step process for each type of analysis for each air speed regime, i.e. subsonic and supersonic as appropriate (fair transonic curve).

Show comparison with known K data values from similar aircraft.

Include a graph of K vs. M

7.6 Drag Polar

Graphs and formulas (place on graph) are required for the following polars:

7.6.1 Configuration CC for subsonic/transonic flight (1 page)

Subcritical M: Re Number based on M = 0.7, 40K Standard
and Critical M through M = 0.95: Intervals of M = 0.05

7.6.2 Configuration CC for supersonic flight (for SSBJ only) (2 page)

Two curves required, one for Endurance Mission and one for High Speed Mission. See SRD for requirements.

7.6.3 Configuration WC (1 page)

Subcritical M: Re Number based on M = 0.6, 30K Standard
and Critical M through M = 0.95: Intervals of M = 0.05

7.6.4 Configuration FC (1 page)

Subcritical M: Re Number based on M = 0.6, 30K Standard
and Critical M through M = 0.95: Intervals of M = 0.05

7.6.5 Configuration WTO (1 page)

Re Number based on V = 100 KTAS, S.L. 90°F day
Gear Down
High lift devices positioned for take-off

7.6.6 Configuration WL (1 page)

Re Number based on V = 100 KTAS, S.L. 90°F day
Hook Down (if applicable)
Gear down and gear up
High lift devices positioned for landing

7.7 One Engine Inoperative (OEI) Drag

7.7.1 Windmilling and Boat Tail Drag

7.7.1.1 Conditions: Reynolds Number for V = 100 KTAS, Sea Level, 90°F day, Configuration: CC

7.7.1.2 Drag shall be calculated and presented. Show sample calculations with step by step process for analysis.

7.7.1.3 Graph Drag (lb) vs. Velocity (knots) on one graph. Use at least 3 points in the range 60 knots to 120 KTAS (i.e. 60, 90, 120)

7.7.2 Trim Drag

7.7.2.1 Conditions: Reynolds Number for V = 100 KTAS, Sea Level, 90°F day, Configuration: CC

7.7.2.2 Drag shall be calculated and presented. Show sample calculations with step by step process for analysis.

7.7.2.3 Graph Drag (lb) vs. Velocity (KTAS) and Drag (lb) vs. Velocity (KCAS). Use at least 3 points in the range 60 knots to 120 KTAS (i.e. 60, 90, 120).

7.8 Moment Coefficients

Determine aircraft C_L vs C_M curve

7.8.1 Conditions: Reynolds Number for V = 100 KTAS, Sea Level, 90°F day, Configuration: CC

7.8.2 Takeoff/landing Curve “Build-up”

7.8.2.1 Show all the details to derive the curves for airfoil, wing, flaps and slats. Document all calculations and adjustments made for twist and incidence and horizontal stabilizer contribution. Construction of curves may be done on graph paper by hand (easiest way).

7.8.3 C_L vs C_M curves for takeoff, climb-out/approach, landing

The graph simplifies and “cleans up” the curves constructed in section 7.8.2, which shall show the C_L vs C_M of the entire aircraft for CC, takeoff and landing configurations.

7.8.4 Cruise

7.8.4.1 See SRD. Flight condition curves will be needed to evaluate various flight conditions, i.e. high-g turns, subsonic and supersonic cruise conditions. For most cases, the linear portion of the curve will be required for performance, stability and control purposes.

7.9 Summary (Standalone page)

Wing planform (reference) area = _____ ft^2

C_L Break = _____ (AoA in deg.)

Critical Mach No. = _____

Three Term Polars, M < Crit.

$$C_D = C_{Do} + _____\ C_L^2 + _____\ (C_L - _____)^2$$

Configuration	Flight Condition	C_{Do}
CC	M=0.7 @40K Standard	
WC	M=0.6 @30K Standard	
WTO	100 knots, sea level 90°, Day; Gear Down, T.O. Flaps	
WL	100 knots, sea level, 90° Day; Gear Down, Speed Brakes, Landing Flaps, Hook Down(if applicable)	

Two Term Polars:

$$C_D = C_{Do} + _____\ C_L^2$$

Flight Condition	C_{Do}		k
	CC	WC	
M<Critical			
M = 0.75 (If applicable)			
M = 0.80 (If applicable)			
M = 0.85			
M = 0.90			
M_{Cruise} (SSBJ only)			

TO/L C_L and C_D

	C_{LMAX}	α for C_{LMAX} (degrees)	C_L @ Ground Attitude	C_D @ Ground Attitude*
TO				
Landing				

*Note C_D @ Ground Attitude includes the reduction in induced drag for ground effect.

8 Propulsion

8.1 Engine Selection

Include rationale and calculations leading to the selection of the engines. Use of engines other than the provided is allowed, but at least two engines shall be used.

Note: Data provided for the F-91 and JT-15 engine assumes installation losses. Any other engine choice requires determination of the installation losses.

If engines are rubberized (i.e. scaled from existing data), show calculations.

8.2 Inlet sizing

8.3 Propulsion Integration Design

Include an 11” x 17” drawing showing the front, top and side view of one installed engine showing the following:

- Inlet design, with area and dimensions
- Nozzle design and dimensions
- Nacelle/pylon configuration (if applicable)

8.4 Performance Data.

Installed performance data for all appropriate conditions in tabular and graphical form shall be included. See F-91 engine data for example of graphs.

9 Performance

The following performance items should be treated in the indicated sequence. For each, present a description of the procedure, a sample calculation, and the required graphs. Only need to present the procedure and sample calculation one time.

Where the area under a curve is used in a calculation, the area and resulting value shall be shown on the graph.

If applicable, show C_{Lbreak} and M_{Cr} on each graph, to show the limitations of the data.

All missions assume Standard Atmosphere, unless otherwise noted.

9.1 Configuration CC- Endurance Mission – **Business Jet only**

9.1.1 Climb

9.1.1.1 One graph each for sea level, 15k, 30k, 36k, 40k, 45k or higher if applicable:

- dh/dt vs. M for at least two weights over the range of possible weights for initial climb
- Each curve should extend to the zero rate of climb axis.

9.1.1.2 Graph Altitude vs. $(dh/dt)_{max}$ for the family of weights. Show ceilings.

9.1.1.3 Graph Altitude vs time, weight, and distance for climb from light off at full fuel

9.1.1.4 Table of Altitude vs time, weight, and distance for climb from light off at full fuel (data supporting section 9.1.1.3)

9.1.2 Cruise (Range and Endurance)

9.1.2.1 One graph each for 15k, 30k, 36k, 40k, 45k or higher if applicable:

- Velocity/Thrust Specific Fuel Consumption (V/TC) vs. M for a family of weights

9.1.2.2 Graph $(V/TC)_{max}$ vs. altitude for the family of weights.

9.1.2.3 Graph $(V/TC)_{max}$ vs. weight for the best range altitude.

9.1.2.4 One graph each for sea level, 15k, 30k, 36k, 40k, 45k or higher if applicable:

- 1/TC vs. M for a family of weights

9.1.2.5 Graph $(1/TC)_{max}$ vs. altitude for the family of weights

9.1.2.6 Graph $(1/TC)_{max}$ vs. weight for best endurance altitude.

9.1.2.7 Graph $(1/TC)_{max}$ vs. weight for sea level, end of mission.

9.1.3 Segment-by-segment profile (graph and table) of design mission.

- Weight, weight of fuel, time, distance, speed and altitude at each transition point.

9.2 Configuration CC- High Speed Mission – **Business Jet only**

9.2.1 Climb and End of Mission

Assume climb and end of mission (after descending) from SRD mission profile is the same as the endurance mission.

9.2.2 Cruise (Range)

9.2.2.1 Graph Velocity/Thrust Specific Fuel Consumption (V/TC) vs. altitude for the family of weights, using required M.

For altitudes, use 15k, 30k, 36k, 40k, 45k or higher if applicable

9.2.2.2 Graph $(V/TC)_{max}$ vs. weight for the best range altitude.

9.2.3 Segment-by-segment profile (graph and table) of the design mission.

- Weight, weight of fuel, time, distance, speed and altitude at each transition point.

9.3 Configuration FC - **CAS and APT only**

9.3.1 Climb

9.3.1.1 One graph each for sea level, 15k, 30k, 36k, 40k, 45k:

- dh/dt vs. M for at least two weights over the range of possible weights for initial climb
- Each curve should extend to the zero rate of climb axis.

9.3.1.2 Graph Altitude vs. $(dh/dt)_{max}$ for the family of weights. Show ceilings.

9.3.1.3 Graph Altitude vs time, weight, and distance for climb from light off at full fuel

9.3.1.4 Table of Altitude vs time, weight, and distance for climb from light off at full fuel (data supporting section 9.3.1.3)

9.3.2 Cruise

9.3.2.1 One graph each for 15k, 30k, 36k, 40k, 45k:

- V/TC vs. M for a family of weights

9.3.2.2 Graph $(V/TC)_{max}$ vs. altitude for the family of weights.

9.3.2.3 Graph $(V/TC)_{max}$ vs. weight for the best range altitude.

9.3.2.4 One graph each for sea level, 15k, 30k, 36k, 40k, 45k:

- 1/TC vs. M for a family of weights

9.3.2.5 Graph $(1/TC)_{max}$ vs. altitude for the family of weights

9.3.2.6 Graph $(1/TC)_{max}$ vs. weight for best endurance altitude.

9.3.2.7 Graph $(1/TC)_{max}$ vs. weight for sea level, end of mission.

9.3.3 Graph segment-by-segment profile (graph and table) of the FC design mission.

- Weight, weight of fuel, time, distance, speed and altitude at each transition point.

9.4 Configuration WC - **CAS and APT only**

9.4.1 Climb

9.4.1.1 One graph each for sea level, 15k, 30k, 36k, 40k, 45k:

- dh/dt vs. M for at least two weights over the range of possible weights for initial climb
- Each curve should extend to the zero rate of climb axis.

9.4.1.2 Graph Altitude vs. $(dh/dt)_{max}$ for the family of weights. Show ceilings.

9.4.1.3 Graph Altitude vs. time, weight and distance for each climb in the design mission

9.4.1.4 Table of Altitude vs time, weight, and distance for climb from light off at full fuel (data supporting section 9.4.1.3)

9.4.2 Cruise

9.4.2.1 Graph Velocity/Thrust Specific Fuel Consumption (V/TC) vs. M for 15k, 30k, 36k, 40k, 45k, for at least three weights over the range of possible weights for the design mission altitude legs

9.4.2.2 Graph $(V/TC)_{max}$ vs. weight for 15k, 30k, 36k, 40k, 45k

9.4.2.3 Graph V/TC vs. M for sea level, at least three weights over the range of weights for the sea level legs of the design mission

9.4.2.4 Graph $(V/TC)_{max}$ vs. weight for the design mission sea level legs

9.4.2.5 Graph (1/TC) vs. M, sea level, for two weights which bracket the reserve fuel requirement for the design mission

9.4.2.6 Graph $(1/TC)_{max}$ vs weight, sea level, end of design mission

9.4.3 Maneuver (See SRD for any overriding requirements)

9.4.3.1 Show sample calculations with step by step process for each type of analysis

9.4.3.2 Graph Max sustained load factor vs. M, 60% fuel, 20k, climb power

9.4.3.3 Graph Turn Radius vs M, 60% fuel, 20K, climb power

9.4.3.4 Graph Turn Rate vs. M, 60% fuel, 20K, climb power

9.4.4 Graph segment-by-segment profile (graph and table) of the WC design mission.

- Weight, weight of fuel, time, distance, speed and altitude at each transition point.

9.5 Takeoff Analysis

Document takeoff analysis showing all work, with assumptions, inputs and outputs. Show takeoff profile in an illustration showing time, distance and speed. Assume sea level, 90°F day, Re for 100 knots. See SRD for details on the requirements.

9.6 Landing Analysis

Document landing analysis showing all work, with assumptions, inputs and outputs. Show landing profile in an illustration showing time, distance and speed. Assume sea level, 90°F day, Re for 100 knots. See SRD for details on the requirements.

9.6.1 Graph Approach speed (knots) vs. weight (lb): full fuel to zero, WL. Use at least three weights

9.6.2 Graph Landing distance (ft) vs. weight (lb): full fuel to zero, WL. Use at least three weights. Include distance for over obstacle, flare and delay time for braking.

9.7 One Engine Inoperative (OEI)

See SRD for details on any specific requirements.

9.7.1 Conditions: Reynolds Number for V = 100 KTAS, Sea Level, 90°F day, Configuration: WTO (CCTO for SSBJ). Assume W_{TO} (maximum ramp weight less 5 minutes at idle)

9.7.2 Show planform view of aircraft, with a freebody diagram, defining all terms for the OEI analysis

9.7.3 Critical Field Length shall be calculated and presented. Show sample calculations with step by step process for analysis.

9.7.4 Graph Velocity (KTAS) vs. Critical Field Length (ft)

9.7.5 Graph OEI Rate of Climb vs. weight: WL, Gear down. Use at least three weights, full fuel to zero.

9.7.6 Graph OEI Rate of Climb vs. weight: WL, Gear up. Use at least three weights, full fuel to zero.

9.8 Performance Summary

9.8.1 Flight Envelope: Altitude vs. M showing stall and buffet limits and thrust limits.

9.8.2 Use the table format of Section 2 to summarize the performance results.

10 Stability and Control

10.1 Static Longitudinal Stability and Control

10.1.1 Present horizontal tail (or canard) sizing showing required static stability criteria are met

10.1.2 Present horizontal tail (or canard) sizing showing required control requirement with the following criteria:

- Trim drag at cruise is <10% total aircraft drag
- High G-maneuver requirement can be met (instantaneous and sustained)
- Take-off rotation can be made at most forward CG location
- High AoA at low speed during landing approach

10.2 Static Directional Stability and Control

10.2.1 Present vertical tail sizing showing required static stability criteria are met.

10.2.2 Present vertical tail sizing showing required control requirement for OEI is met.

10.3 Aileron Sizing **(AE460B only)**

Present aileron sizing to ensure the roll requirement is met (see SRD)

10.4 Stability of Uncontrolled Motion, Dynamic Stability **(AE460B only)**

The section shall include all inputs to the non-dimensional stability derivatives, including flight condition being analyzed, aircraft design properties, i.e. weight, CG, wing, fuselage and tail dimensions, aerodynamic properties, and steady state coefficients. Include a summary table showing the weight, altitude, Mach No., angle of attack, plus Ixx, Iyy, Izz and Ixz about the stability axes.

See class presentation for information required in this report for diagrams, equations and sample calculations.

Computer code and/or spreadsheets shall be submitted to the instructor electronically (to save paper) or included in the appendix.

Results shall be presented similar as shown in class and provided on course website.

11 Configuration Spreadsheet

A standardized spreadsheet will be provided by the instructor to submit your configuration. Do not reformat the spreadsheet. Ensure values are provided in the specified units.

12 Life Cycle Costs

Include Development, Test and Evaluation (DT&E) costs in this year's dollars for 9 development aircraft with a production order of 200. AIAA competition designs should follow any quantities given in the AIAA instructions. See Nicolai, CH 24 for table format.

13 References

In all cases references shall be cited.

14 Bibliography

Add any published works used that are not cited in the report, i.e. military specifications, NACA reports, instruction manuals for software programs, etc.

Appendices

A. Work Breakdown Structure (signed)

B. Computer Program Listings

   Source Code Listings for ALL programs other than the products of major software such as EXCEL, WORD, etc. used, shall be included. May supply the code on a memory stick. If supplied on disk, describe the code in this section that correlates to your code. Index the code with F.n (where n is the index for each code, i.e. F.1., F.2., etc.).

C. EXCEL spreadsheets used for input to report. May supply the spreadsheets on a memory stick. If supplied on a memory stick, describe the code in this section that correlates to your code. Index the spreadsheets with G.n (where n is the index for each spreadsheet, i.e. G.1., G.2., etc.).